DAY 25 – BASIC INCIDENT RESPONSE

Cyber Security Internship – Veda Technology

1. Objective

To understand the fundamental stages of cybersecurity incident response and develop a structured checklist for identifying, managing, containing, and recovering from security incidents. The task focuses on understanding how organizations respond to security threats and minimize potential damage.

2. Introduction

Incident response is a systematic process used by organizations to identify, investigate, contain, and recover from cybersecurity incidents.

Security incidents may include malware infections, unauthorized access, phishing attacks, data breaches, and suspicious network activities.

An effective incident response process helps organizations reduce the impact of cyberattacks, protect sensitive information, restore normal operations, and prevent similar incidents in the future.

Incident response requires proper planning, communication, documentation, and coordination between security teams and system administrators.

3. Common Types of Security Incidents

Incident Type	Description	Possible Impact
Malware Infection	Malicious software infects a system	Data loss or system disruption
Phishing Attack	Fraudulent communication attempts to steal information	Credential theft
Unauthorized Access	An unknown or unauthorized user accesses resources	Data exposure
Ransomware Attack	Malicious software encrypts files and demands payment	Operational disruption
Data Breach	Sensitive information is accessed or disclosed without authorization	Privacy and financial risks
Denial-of-Service Attack	Excessive traffic disrupts a service	Service unavailability

4. Stages of Incident Response

The incident response process consists of several important stages.

Stage 1: Preparation

Preparation involves establishing policies, tools, procedures, and responsibilities before a security incident occurs.

Key Activities:

- Develop an incident response plan.
- Assign responsibilities to security team members.
- Maintain updated security tools.
- Conduct cybersecurity awareness training.
- Prepare secure data backups.
- Establish communication and escalation procedures.

Expected Outcome: The organization is prepared to respond effectively when an incident occurs.

Stage 2: Detection and Analysis

This stage focuses on identifying suspicious activities and determining whether a security incident has occurred.

Key Activities:

- Monitor security alerts and system logs.
- Identify unusual login attempts.
- Analyze suspicious network traffic.
- Examine potential malware alerts.
- Determine affected systems and accounts.
- Assess the severity and scope of the incident.

Expected Outcome: Potential incidents are identified, analyzed, and prioritized for investigation.

Stage 3: Containment

Containment involves limiting the spread of a security incident and preventing additional damage.

Key Activities:

- Isolate affected devices when appropriate.
- Restrict compromised user accounts.
- Block confirmed malicious connections.
- Prevent unauthorized access to sensitive resources.

- Preserve relevant evidence before making major changes where feasible.
- Apply temporary security restrictions.

Expected Outcome: The incident is controlled and further damage is minimized.

Stage 4: Eradication

Eradication focuses on eliminating the root cause of the incident and removing malicious components.

Key Activities:

- Remove detected malware.
- Eliminate unauthorized accounts or access mechanisms.
- Correct exploited vulnerabilities.
- Apply necessary security patches.
- Reset compromised credentials.
- Verify that the identified threat has been removed.

Expected Outcome: The known cause of the incident is eliminated from affected systems.

Stage 5: Recovery

Recovery involves restoring affected systems and services to normal operation while ensuring that security risks have been addressed.

Key Activities:

- Restore systems from verified backups when necessary.
- Validate system functionality.
- Verify the integrity of recovered data.
- Monitor systems for recurring suspicious activity.
- Reconnect isolated devices when considered safe.
- Confirm that business operations are functioning normally.

Expected Outcome: Systems are safely restored and monitored for signs of reinfection or repeated compromise.

Stage 6: Lessons Learned

This stage involves reviewing the incident to identify weaknesses and improve future security responses.

Key Activities:

- Document the incident timeline.
- Review the effectiveness of response actions.
- Identify the root cause and contributing factors.
- Update security policies and procedures.
- Recommend preventive security controls.
- Provide additional security awareness training where necessary.

Expected Outcome: The organization improves its security practices and incident response readiness.

5. Incident Response Process Summary

Stage	Main Purpose	Key Action
Preparation	Establish response readiness	Create policies and procedures
Detection and Analysis	Identify and assess incidents	Monitor alerts and logs
Containment	Limit further damage	Isolate affected resources
Eradication	Remove the cause	Eliminate threats and vulnerabilities
Recovery	Restore normal operations	Verify and restore systems
Lessons Learned	Improve future responses	Review and document findings

6. Practical Scenario – Suspicious Login Activity

A hypothetical organization detects repeated failed login attempts targeting an administrator account.

The security monitoring system identifies several authentication failures originating from an unfamiliar source IP address.

The incident response team follows a structured approach to investigate and address the activity.

Incident Details

Parameter	Description
Incident ID	IR-025
Incident Type	Suspicious Authentication Activity
Detection Method	Authentication Log Monitoring

Affected Resource	Administrator Account
Observed Activity	Multiple failed login attempts
Initial Severity	Medium
Incident Status	Simulated – Under Investigation

Incident Response Actions

Stage	Response Action
Preparation	Review established authentication security procedures
Detection	Identify repeated failed login attempts
Analysis	Examine timestamps, usernames, and source IP addresses
Containment	Apply appropriate rate limits or temporary access restrictions
Eradication	Address any confirmed compromise or exploited weakness
Recovery	Verify authorized access and restore normal operations
Lessons Learned	Document findings and improve authentication monitoring

Observation:

Repeated login failures may indicate password guessing or unauthorized access attempts. However, additional evidence is required to confirm malicious activity.

7. Incident Response Checklist

The following checklist can be used to guide the handling of basic cybersecurity incidents.

A. Preparation

- Establish an incident response plan.
- Identify responsible security personnel.
- Maintain updated security tools.
- Prepare secure and verified backups.
- Define incident communication procedures.

B. Detection and Analysis

- Identify the security alert or suspicious activity.
- Record the date and time of detection.
- Examine relevant security logs.
- Identify potentially affected systems.

- Determine the incident severity.
- Document initial findings.

C. Containment

- Isolate affected systems when necessary.
- Restrict compromised accounts.
- Block confirmed malicious activity.
- Preserve relevant logs and evidence.
- Monitor for additional suspicious events.

D. Eradication

- Identify the root cause.
- Remove malicious software or unauthorized access.
- Apply required security patches.
- Reset compromised credentials.
- Verify that identified threats are removed.

E. Recovery

- Restore affected systems safely.
- Verify system functionality.
- Check the integrity of restored data.
- Monitor for recurring threats.
- Confirm normal operations.

F. Lessons Learned

- Prepare an incident summary.
- Document the incident timeline.
- Evaluate the effectiveness of response actions.
- Identify security improvements.
- Update incident response procedures.

8. Incident Severity Classification

Security incidents are prioritized based on their potential impact and urgency.

Severity	Description	Example	Response Priority
Low	Minor event with limited impact	Isolated unsuccessful login	Routine review
Medium	Suspicious activity requiring investigation	Repeated login failures	Timely investigation
High	Serious incident affecting important resources	Confirmed unauthorized account access	Urgent response
Critical	Major incident affecting essential systems	Widespread ransomware disruption	Immediate escalation

Severity depends on the available evidence, affected assets, business impact, and organizational policies.

9. Importance of Incident Response

Incident response plays a significant role in maintaining organizational cybersecurity.

1. Damage Reduction: Helps limit the spread and impact of security incidents.
2. Faster Recovery: Supports the restoration of affected systems and services.
3. Data Protection: Reduces the risk of unauthorized access and information loss.
4. Threat Investigation: Helps identify attack methods and affected resources.
5. Operational Continuity: Minimizes disruption to important services.
6. Security Improvement: Provides lessons that strengthen future defenses.

10. Best Practices for Effective Incident Response

- Maintain a documented incident response plan.
- Continuously monitor security events and alerts.
- Establish clear communication and escalation procedures.
- Preserve incident evidence for investigation.
- Follow the principle of least privilege.
- Regularly update operating systems and security software.
- Maintain secure backups and test recovery procedures.
- Conduct periodic incident response exercises.
- Review and improve response procedures after incidents.

11. Results and Observations

The theoretical study and simulated incident scenario demonstrated how organizations can respond to cybersecurity incidents using a structured process.

Activity	Result
Incident Response Study	Six response stages reviewed
Incident Identification	Suspicious login activity scenario examined
Incident Analysis	Potential causes and risks identified
Containment Planning	Appropriate access restrictions considered
Recovery Planning	Safe restoration and verification steps outlined
Checklist Development	Stage-wise response checklist prepared
Security Recommendations	Preventive and corrective measures documented

Key Observations

- Early detection helps reduce the potential impact of security incidents.
- Containment prevents incidents from spreading to additional systems.
- Eradication addresses the underlying cause of confirmed security threats.
- Recovery requires validation before normal operations resume.
- Documentation and lessons learned improve future incident response capabilities.

12. Conclusion

The Basic Incident Response task provided an understanding of the essential stages involved in managing cybersecurity incidents.

The study covered preparation, detection and analysis, containment, eradication, recovery, and lessons learned. A simulated suspicious login scenario was used to demonstrate how these stages can be applied to a potential security incident.

A structured incident response checklist was also developed to guide security teams through identifying, investigating, controlling, and recovering from incidents.

Overall, this task strengthened foundational knowledge of cybersecurity incident management and highlighted the importance of preparation, timely detection, effective containment, and continuous security improvement.